

Task 7: UART

1. UART ESP32 To PC

```
7_uart.ino
1  // ESP32 UART Communication Example
2  // This sends and receives data via Serial Monitor
3
4  void setup() {
5      Serial.begin(115200); // Start serial communication with laptop
6      Serial.println("ESP32 UART Communication Started!");
7  }
8
9  void loop() {
10     // Send a message every second
11     Serial.println("Msg from ESP32!");
12     delay(1000);
13
14     // If data is received from Serial Monitor, print it back
15     if (Serial.available()) {
16         String msg = Serial.readStringUntil('\n');
17         Serial.print("Received: ");
18         Serial.println(msg);
19     }
20 }
21
```

2. UART ESP32 To ESP32

Sender Code

uart_sender.ino

```
1 // Define TX and RX pins for UART (change if needed)
2 #define TXD1 19
3 #define RXD1 21
4
5 // Use Serial1 for UART communication
6 HardwareSerial mySerial(1);
7
8 int counter = 0;
9
10 void setup() {
11     Serial.begin(115200);
12     mySerial.begin(9600, SERIAL_8N1, RXD1, TXD1); // UART setup
13
14     Serial.println("ESP32 UART Transmitter");
15 }
16
17 void loop() {
18
19     // Send message over UART
20     mySerial.println(String(counter));
21
22     Serial.println("Sent: " + String(counter));
23
24     // increment the counter
25     counter++;
26
27     delay(1000);
28 }
```

Receiver Code

uart_receiver.ino

```
1 // Define TX and RX pins for UART (change if needed)
2 #define TXD1 19
3 #define RXD1 21
4
5 // Use Serial1 for UART communication
6 HardwareSerial mySerial(2);
7
8 void setup() {
9     Serial.begin(115200);
10    mySerial.begin(9600, SERIAL_8N1, RXD1, TXD1); // UART setup
11
12    Serial.println("ESP32 UART Receiver");
13 }
14
15 void loop() {
16     // Check if data is available to read
17     if (mySerial.available()) {
18         // Read data and display it
19         String message = mySerial.readStringUntil('\n');
20         Serial.println("Received: " + message);
21     }
22 }
23
```